Satellite Video Continuous Space–Time Super-Resolution via Mask-Based Temporal-Aware Warping and Cross-Level Frequency Integration

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***Abstract*—The rapid advancement of remote sensing technology has driven a growing demand for high-quality satellite video. However, constrained by transmission bandwidth and storage costs, satellite videos are typically stored and transmitted at low spatial resolutions and frame rates. As a key solution, space–time super-resolution (STSR) aims to enhance video quality. Unfortunately, existing satellite video STSR methods fail to effectively model interframe continuity and, consequently, only support a fixed upsampling scale, which restricts their flexibility and practical applicability. To address this challenge, we propose a novel continuous STSR (CSTSR) framework for satellite videos, termed SV-CSTSR. Specifically, first, we develop a spatial–frequency joint modulation block (SFJMB), which aims to jointly mine the fine-grained features of satellite videos in the spatial and frequency domains by using a dual-branch structure. Second, we propose a mask-based temporal-aware warping module (MTWM) to model continuous feature representations in the temporal dimension by modulating feature warping with a time factor, enabling frame interpolation at arbitrary times. Third, we design an interframe deformable attention module (IDAM), which can explore interframe dependencies to achieve effective information fusion by adaptively selecting the positions of key and value pairs in a data-dependent manner. Finally, we propose a cross-level frequency integration module (CFIM) for continuous-scale upsampling. CFIM aims to activate channel responses through frequency selection (FS), thereby achieving adaptive integration of cross-level frequency latent codes to learn feature representations of satellite videos at arbitrary resolution. Extensive experiments demonstrate that the proposed SV-CSTSR surpasses state-of-the-art (SOTA) methods, achieving superior quantitative accuracy and perceptual fidelity.**

***Index Terms*—Continuous-scale upsampling, satellite video, space–time super-resolution (STSR), spatial–frequency joint modulation, temporal-aware warping.**

I. Introduction

With the increase in the number of satellite launches, satellite video, as a novel data source, is being used in many fields, such as target tracking [1], [2], [3], object segmentation [4], [5], [6], and moving vehicle detection [7], [8], [9]. Currently, the rapid development of computer vision technologies has underscored the growing need for high-quality satellite videos to enable reliable and fine-grained visual analysis. However, on the one hand, due to the limitations of satellite video sensors and the high observation distance of satellites to the Earth, the spatial resolution of satellite videos is generally low; on the other hand, due to the limitations of transmission bandwidth and storage costs, satellite videos usually sacrifices its spatial and temporal resolution, which further reduces the quality of satellite videos. Consequently, restoring low-quality satellite videos to high spatial and temporal resolution has become an urgent and critical challenge, which is also the key to meeting downstream high-precision computer vision tasks. Although existing enhancement methods [10], [11], [12], [13], [14] have achieved remarkable progress, they cannot be directly applied to satellite videos to simultaneously improve both spatial and temporal resolutions.

Satellite video space–time super-resolution (STSR) aims to simultaneously improve the spatial resolution and frame rate (temporal resolution) of satellite videos, which can effectively handle the above urgent challenge. However, most existing satellite video super-resolution (SR) methods [15], [16], [17], [18], [19], [20] focus solely on spatial resolution and cannot achieve satellite video temporal SR. To enable these methods with temporal SR capability, a straightforward strategy is to combine them with video frame interpolation (VFI) techniques. Nevertheless, such a two-stage pipeline based on such a combination often leads to cumulative errors between networks, resulting in suboptimal STSR performance. Instead of performing SR in space and time separately in two stages, this work [21] performs spatial and temporal SR jointly within a single stage, effectively filling the gap in satellite video STSR research. Unfortunately, progress in this field has largely stagnated over the past three years. Moreover, the method in [21] is designed for satellite video fixed-scale STSR (FSSTSR), which restricts the reconstruction to a predetermined spatial and temporal upsampling ratio, as illustrated in Fig. 1(a). While satellite video FSSTSR demonstrates promising reconstruction results, the fixed space and time scale ratio is not conducive to the flexible presentation of satellite videos, which will limit its application scenarios and practical deployment. Because in real-world satellite video applications, display devices often vary in spatial resolution and screen refresh rate, making it desirable to visualize content at different spatial and temporal resolutions. Furthermore, during interactive monitoring, users frequently perform continuous zooming

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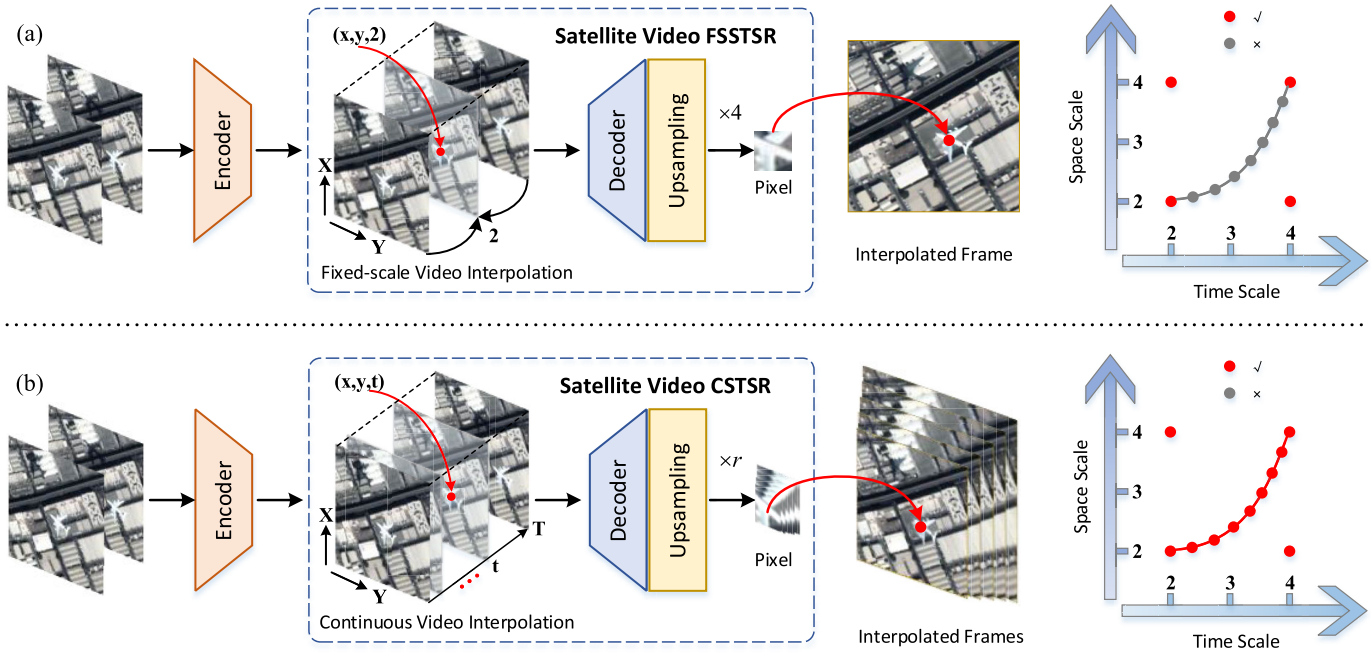


Fig. 1. Comparison of implementation processes and results between (a) FSSTSR and (b) CSTSR. ✓ (red circle) indicates that the corresponding function is supported, while × (gray circle) denotes that it is not.

operations (e.g., via mouse scroll), underscoring the necessity for continuous and adaptive resolution scaling in both space and time.

In this article, as illustrated in Fig. 1(b), to break through the limitation of FSSTSR, we propose a unified framework SV-CSTSR to achieve satellite video continuous STSR (CSTSR). Specifically, first, considering that the texture of satellite video is weak and the detailed features are difficult to extract, we develop a spatial–frequency joint modulation block (SFJMB). SFJMB employs a dual-branch structure to collaboratively mine the detailed features of satellite videos in both the spatial domain and frequency domain, where the spatial branch focuses on local feature extraction, while the frequency branch models global dependencies and adaptively modulates high-frequency (detail) and low-frequency (structure) components, effectively dealing with the weak texture of satellite videos. Second, to effectively learn the continuous feature representation of satellite video in the temporal dimension, we propose a mask-based temporal-aware warping module (MTWM) to model the signal along a continuous temporal axis between two consecutive input frames. Specifically, MTWM first exploits interframe differences to focus on motion regions, enabling more accurate estimation of interframe motion offsets. The offsets are modulated by a continuous time factor and then fed into deformable convolution [22], [23] to perform adaptive feature warping, thereby enabling video decoding at arbitrary frame rates. Third, we design an interframe deformable attention module (IDAM) to facilitate interframe feature interaction. Interframe deformable attention (IDA) can adaptively select the positions of key and value pairs in a data-dependent manner, which can help capture feature dependencies between frames to better utilize interframe complementary information. Additionally, we embedded a texture enhancement block (TEBlock) in IDAM to improve the local representation of weak features in satellite videos. Finally, we propose a cross-level frequency integration module (CFIM) to model the continuous representation of features in coordinates, thereby enabling continuous-scale upsampling. CFIM first performs frequency selection (FS) on multiresolution feature maps to enhance key information, and then integrates cross-level frequency latent codes based on channel responses under the conditioning of continuous spatial coordinates, enabling the network to learn optimal feature mappings in a continuous-scale space.

The main contributions of this article are summarized as follows.

1) We propose a novel framework for satellite video CSTSR. Our SV-CSTSR enables continuous modeling of satellite videos in both the spatial and temporal dimensions, thereby learning efficient feature representations at flexible resolutions and decoding videos to arbitrary spatial resolutions and frame rates.
2) We develop an SFJMB. Our SFJMB integrates spatial modulation for fine-grained local texture modeling and frequency modulation (FM) for capturing global feature dependencies, enabling effective feature extraction from satellite videos.
3) We design an MTWM and an IDAM. MTWM can learn continuous modeling of satellite video in the temporal dimension, achieving frame interpolation at any time. IDAM can learn interframe feature dependencies to facilitate the effective aggregation of complementary information.
4) We propose a CFIM for continuous-scale upsampling. CFIM can model the spatial continuity of features and employs FS to activate channel responses for the

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adaptive integration of cross-level frequency latent codes, enabling optimal continuous-scale upsampling.

II. Related Works

In this section, we first provide a brief review of VFI, then present an overview of satellite video SR (VSR), and finally summarize existing studies on video STSR.

A. Video Frame Interpolation

VFI [24], [25], [26], [27] is a technology that generates intermediate frames between two adjacent frames, aiming to improve the temporal resolution (frame rate) of videos. The key to VFI is how to warp the pixels/features of the reference frame to the interpolation frame based on motion information. Most VFI methods [28], [29], [30], [31], [32] usually use an optical flow network to predict motion information. For example, Memc-net [28] developed a novel adaptive warping layer to integrate both optical flow and interpolation kernels to predict target frame pixels, Xu et al. [31] proposed a quadratic video interpolation method that allows for the prediction of curved trajectories and variable-speed motion, generating more accurate interpolation results. However, optical flow calculation requires huge computational costs and is time-consuming, which cannot meet the real-time requirements of video processing. Therefore, many methods [33], [34], [35] based on deformable convolution [22] have been proposed. Cheng and Chen [33] proposed to use deformable separable convolution to adaptively estimate kernels, offsets, and masks, enabling the network to capture information from fewer yet more relevant pixels while delivering strong performance. Recently, with the development of image processing technology, Shi et al. [36] proposed a Transformer-based VFI framework, which aims to consider long-range dependencies with the self-attention mechanism. Zhang et al. [27] proposed VFIMamba, a novel VFI method for efficient and dynamic interframe modeling by using state-space models. Danier et al. [26] proposed a latent diffusion model-based VFI, named LDMVFI. LDMVFI regards the VFI task as a conditional generation problem from a generative perspective and can interpolate video content with favorable perceptual quality.

B. Satellite VSR

Recent years have witnessed significant advancements in satellite VSR. The field was pioneered by Liu et al. [15], who established the first effective satellite VSR framework incorporating local spatiotemporal neighbors and nonlocal similarity modeling. Then, Xiao et al. [16] proposed a novel fusion strategy of temporal grouping projection with deformable convolution to improve the fusion effect of spatiotemporal information. At the same time, Liu and Gu [37] proposed a deep joint estimation network to deal with the problem that the blur kernel is usually unknown. After that, Jin et al. [38] proposed a novel two-branch alignment network, which aims to improve the ability to model long-distance correspondence by deformable convolution and Transformer-like attention mechanism. Recently, Xiao et al. [39] proposed a practical blind satellite video super-resolution (SVSR) method to explore sharp textures by considering the pixel-wise blur levels in a coarse-to-fine manner. The above satellite VSR methods limit the upsampling scales to specific integer values and require training a distinct model for each upsampling scale. To solve the fixed-scale limitation, Ni and Zhang [40] proposed a novel network for continuous-scale satellite VSR, which achieved good performance on both fixed-scale factors and continuous-scale factors. Recently, to learn more efficient motion cues, Wang et al. [41] proposed a phase-based satellite VSR method, PhaseVSRnet, which aims to enlarge the motion information with phase shifts by converting satellite video frames into the phase domain. The results show that PhaseVSRnet effectively captures motion cues in the phase domain and obtains a strong generalization capability.

Although satellite VSR has made significant progress, most methods can only achieve spatial SR and cannot improve the temporal resolution of satellite videos. In this article, we propose the MTWM, which can accurately learn the motion offset of features in the temporal dimension, thereby completing VFI at an arbitrary time.

C. Space–Time SR

STSR aims to simultaneously increase the spatial and temporal resolutions of the given low-resolution (LR) and low frame rate videos.

In the field of natural videos, the technology of STSR is mature, and many methods [42], [43], [44], [45] have been proposed before the popularization of deep learning (DL). In recent years, with the rapid development of DL, some methods [46], [47], [48], [49], [50] employ powerful network fitting capabilities to address the task. Geng et al. [49] proposed to use space–time Transformers to naturally merge spatial and temporal SR modules into a single model to solve the slow inference speed. Wang et al. [51] proposed a novel bidirectional recurrent space–time upsampling network to explore and utilize the auxiliary information from adjacent time steps. In order to make VFI more flexible, Xu et al. [48] proposed a temporal modulation block, which allows STSR to be continuous in the time scale. Inspired by this work [48], in the following years, researchers have devoted themselves to the study of continuous SR and proposed many CSTSR methods [52], [53], [54], [55], [56]. Chen et al. [52] proposed the first CSTSR network, which allows continuous SR on both space and time scales. Zhang et al. [55] designed a Galerkin-type attention function to perform frame alignment and temporal interpolation to complete efficient and accurate motion estimation and motion compensation.

In the field of satellite videos, most existing satellite VSR methods aim to improve the spatial resolution and ignore the temporal resolution. To solve this limitation, Xiao et al. [21] proposed a joint framework to enhance the spatial and temporal resolution of satellite video simultaneously. Although this work [21] has achieved good performance, it can only perform SR on predefined fixed-scale factors, which makes the application scope limited. SV-CSTSR can perform continuous SR in both spatial and temporal dimensions, which is more flexible in application and will also broaden the application scope of satellite video.

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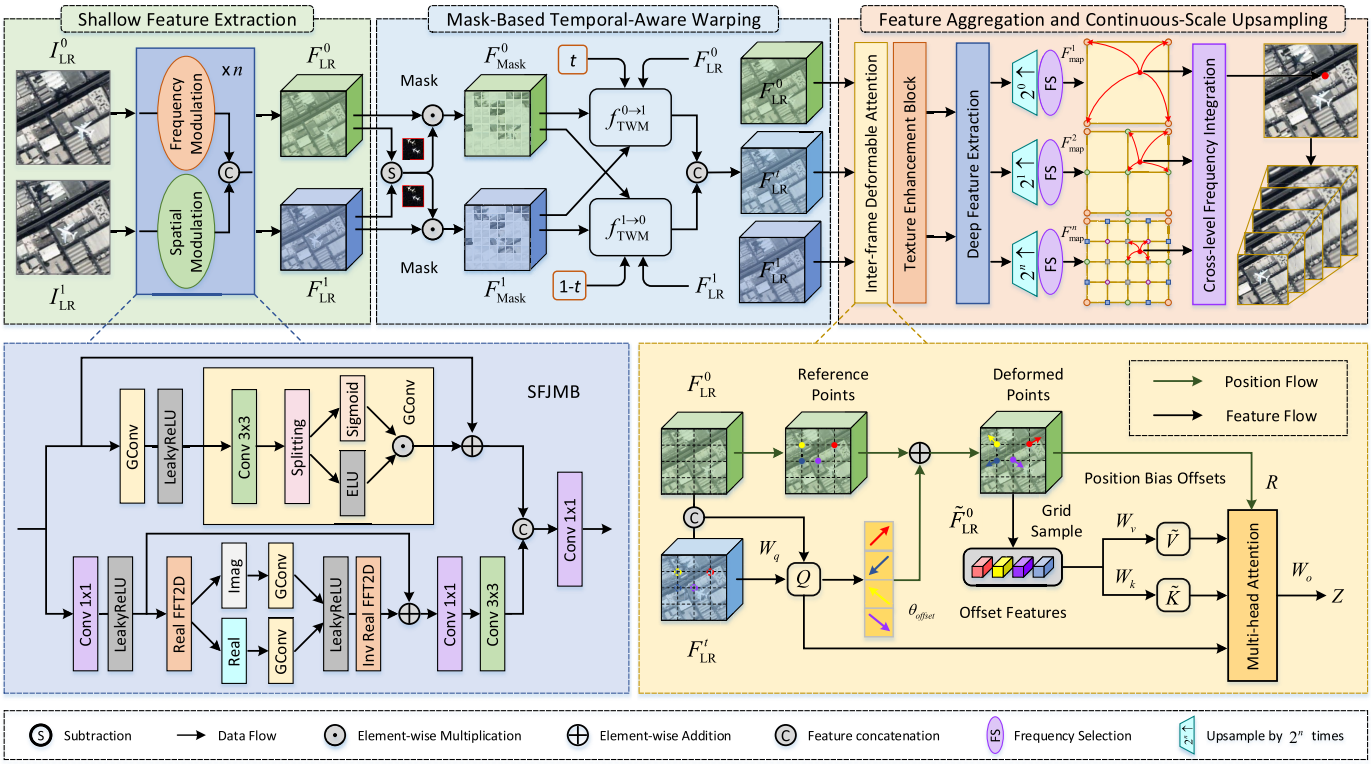


Fig. 2. Overall architecture of the proposed SV-CSTSR and structure of SFJMB.

III. Proposed Method

First, we provide an overview of our proposed SV-CSTSR in Section III-A. Then, we introduce the proposed components SFJMB, MTWM, IDAM, and CFIM in Sections III-B–III-E, respectively. Finally, in Section III-F, we provide a detailed description of the loss functions.

A. Overview

As shown in Fig. 2, our SV-CSTSR aims to use a unified network to achieve CSTSR for satellite videos. In Fig. 2, the annotation "Data Flow" refers to the sequential propagation path of features or video frame data in our network; "Subtraction," "Element-wise Multiplication," and "Element-wise Addition" represent element-wise subtraction, multiplication, and addition operations, respectively. In addition, the annotation "Feature Concatenation" denotes the concatenation of features along the channel dimension. The annotation "FS" denotes the operation of selecting and enhancing specific frequency components from the input, and "Upsample by 2^n times" denotes the operation of upsampling the feature maps by a factor of 2^n. The proposed SV-CSTSR consists of three parts: shallow feature extraction, mask-based temporal-aware warping, and feature aggregation and continuous-scale upsampling.

Specifically, given two consecutive video frames $\boldsymbol{I}^0_{\mathrm{LR}}$ and $\boldsymbol{I}^1_{\mathrm{LR}}$, the process proceeds as follows.

First, in the shallow feature extraction stage, we employ a series of k SFJMBs to extract shallow features, yielding features $\boldsymbol{F}^0_{\mathrm{LR}}$ and $\boldsymbol{F}^1_{\mathrm{LR}}$. These features serve as the initial descriptive embedding of the input frame, providing essential structural and texture details for the subsequent MTWM.

Second, in the mask-based temporal-aware warping stage, suppose that we want to predict the video frame at arbitrary time t between the two video frames $\boldsymbol{I}^0_{\mathrm{LR}}$ and $\boldsymbol{I}^1_{\mathrm{LR}}$, and we use MTWM to perform adaptive feature warping on $\boldsymbol{F}^0_{\mathrm{LR}}$ and $\boldsymbol{F}^1_{\mathrm{LR}}$ to generate the intermediate video frame features $\boldsymbol{F}^t_{\mathrm{LR}}$. Here, $\boldsymbol{F}^t_{\mathrm{LR}}$ denotes a coarse predicted feature at time t.

Third, in the feature aggregation and continuous-scale upsampling stage, we first employ the proposed IDAM to model interframe dependencies, thereby facilitating the effective aggregation of complementary information across frames. Subsequently, a series of K SFJMBs is used for deep feature mining, ensuring the preservation of both local details and global context.

Finally, the enriched feature representation is passed to CFIM, which performs continuous-scale upsampling via cross-level FS to generate high-resolution (HR) outputs $\boldsymbol{I}^t_{\mathrm{SR}}$ at arbitrary magnification factors.

B. Spatial–Frequency Joint Modulation Block

As shown in Fig. 2, our SFJMB includes two parallel branches: a spatial modulation branch and an FM branch. Suppose that the input feature is denoted as $\boldsymbol{X}_{\mathrm{in}}$. In the spatial modulation branch, we use two layers of convolution to capture local spatial dependencies and enhance structural details. It is worth noting that we chose to use gated convolution [57] here, which can learn a dynamic feature selection according to the input content, thereby effectively suppressing invalid or interfering features and retaining key features,

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thereby enabling more effective feature extraction. The specific operations are as follows:

$$\boldsymbol{X}_{\mathrm{S}} = \mathtt{GConv}_{3\times3}\left(\mathtt{LeakyReLU}\left(\mathtt{GConv}_{3\times3}\left(\boldsymbol{X}_{\mathrm{in}}\right)\right)\right) \tag{1}$$

where $\boldsymbol{X}_{\mathrm{S}}$ denotes the output of the spatial modulation branch, $\mathtt{GConv}_{3\times3}(\cdot)$ represents the operations of the 3×3 gated convolution, and $\mathtt{LeakyReLU}(\cdot)$ is the activate function. In the FM branch, we first use 1×1 convolution to perform feature embedding, formulated as follows:

$$\boldsymbol{X}_{\mathrm{E}} = \mathtt{LeakyReLU}\left(\mathtt{Conv}_{1\times1}\left(\boldsymbol{X}_{\mathrm{in}}\right)\right). \tag{2}$$

Subsequently, we apply a 2-D fast Fourier transform (FFT) to transform the embedded feature $\boldsymbol{X}_{\mathrm{E}}$ into the frequency domain for FM, thereby obtaining a global receptive field. Moreover, we separate the real part and the imaginary part during modulation to retain their independence, which is conducive to fine-grained feature modeling and makes feature extraction more suitable for data with strong feature differences and complex structures, such as satellite videos. The operation is defined as follows:

$$\left[\boldsymbol{X}_{\mathrm{real}}, \boldsymbol{X}_{\mathrm{imag}}\right] = \mathtt{FFT}\left(\boldsymbol{X}_{\mathrm{E}}\right) \tag{3}$$

$$\begin{aligned}\hat{\boldsymbol{X}}_{\mathrm{real}} &= \mathtt{GConv}_{1\times1}\left(\boldsymbol{X}_{\mathrm{real}}\right), \hat{\boldsymbol{X}}_{\mathrm{imag}} \\ &= \mathtt{GConv}_{1\times1}\left(\boldsymbol{X}_{\mathrm{imag}}\right)\end{aligned} \tag{4}$$

where $\mathtt{FFT}(\cdot)$ denotes the 2-D FFT function and $\boldsymbol{X}_{\mathrm{real}}$ and $\boldsymbol{X}_{\mathrm{imag}}$ represent the real part and imaginary part after the 2-D FFT, respectively. After that, we perform the inverse FFT to obtain the modulated features as follows:

$$\hat{\boldsymbol{X}}_{\mathrm{E}} = \mathtt{LeakyReLU}\left(\mathtt{IFFT}\left(\left[\hat{\boldsymbol{X}}_{\mathrm{real}}, \hat{\boldsymbol{X}}_{\mathrm{imag}}\right]\right)\right) \tag{5}$$

where $\mathtt{IFFT}(\cdot)$ denotes the inverse FFT operation. Here, to prevent potential gradient explosion, we introduce a residual connection before performing feature de-embedding, which is implemented using a 1×1 convolution followed by a 3×3 convolution. Finally, the outputs from the two branches are fused to generate the final output $\boldsymbol{X}_{\mathrm{out}}$ of the SFJMB as follows:

$$\boldsymbol{X}_{\mathrm{F}} = \mathtt{Conv}_{3\times3}\left(\mathtt{Conv}_{1\times1}\left(\hat{\boldsymbol{X}}_{\mathrm{E}} + \boldsymbol{X}_{\mathrm{E}}\right)\right) \tag{6}$$

$$\boldsymbol{X}_{\mathrm{out}} = \mathtt{Conv}_{1\times1}\left(\mathbb{C}\left[\boldsymbol{X}_{\mathrm{S}}, \boldsymbol{X}_{\mathrm{F}}\right]\right) \tag{7}$$

where $\boldsymbol{X}_{\mathrm{F}}$ represents the output of the frequency branch and $\mathbb{C}[\cdot,\cdot]$ represents the feature concatenation operation.

C. Mask-Based Temporal-Aware Warping Module

To achieve continuous VFI, we model the video signal along a continuous temporal axis between two consecutive input frames. Given two adjacent frames, the temporal interval between them is normalized to $[0, 1]$, where each intermediate time instant $t \in [0, 1]$ corresponds to a potential interpolated frame. Instead of predicting frames only at fixed discrete time steps, the proposed model learns a continuous temporal representation that allows frame synthesis at arbitrary temporal positions. To enable our SV-CSTSR to achieve this capability, we propose MTWM. Specifically, given two adjacent frames features $\boldsymbol{F}^0_{\mathrm{LR}}$ and $\boldsymbol{F}^1_{\mathrm{LR}}$ at time indices 0 and 1, the intermediate time is represented by a continuous variable $t \in [0, 1]$. MTWM predicts the intermediate video frames by conditioning on the continuous temporal parameter t, which can be formulated as

$$\boldsymbol{F}^t_{\mathrm{LR}} = \mathcal{W}\left(\boldsymbol{F}^0_{\mathrm{LR}}, \boldsymbol{F}^1_{\mathrm{LR}}; t\right) \tag{8}$$

where $\mathcal{W}(\cdot)$ denotes the operations of the proposed MTWM. MTWM aims to adjust the offset of the deformable convolution according to a given time factor t to complete the required feature warping. Specifically, we perform feature warping for frame interpolation by estimating the forward motion clues from $\boldsymbol{I}^0_{\mathrm{LR}}$ to intermediate frame $\boldsymbol{I}^t_{\mathrm{LR}}$ and the backward motion clues from $\boldsymbol{I}^1_{\mathrm{LR}}$ to intermediate frame $\boldsymbol{I}^t_{\mathrm{LR}}$. Due to the significant scale variations of moving targets and the frequent occurrence of small objects in satellite videos, it is difficult to capture motion cues effectively. Therefore, inspired by the pyramid structure [23], [48], we apply deformable convolution on multilevel solution feature maps to better perceive motion clues. In addition, we introduce an effective mask strategy, which aims to use the difference of $\boldsymbol{F}^0_{\mathrm{LR}}$ and $\boldsymbol{F}^1_{\mathrm{LR}}$ to distinguish between motion offset areas and nonmotion offset areas. We only predict offsets of deformable convolution in the motion offset area to complete feature warping, which makes the network focus on the motion targets and thereby improves the accuracy of offset prediction. The detailed process of MTWM is described as follows.

First, we compute the difference feature map M_{diff} by subtracting $\boldsymbol{F}^0_{\mathrm{LR}}$ from $\boldsymbol{F}^1_{\mathrm{LR}}$, as formulated in the following:

$$\boldsymbol{M}_{\mathrm{diff}} = \boldsymbol{F}^0_{\mathrm{LR}} - \boldsymbol{F}^1_{\mathrm{LR}}. \tag{9}$$

Then, we transform $\boldsymbol{M}_{\mathrm{diff}}$ into a motion area mask $\boldsymbol{M}$ by the following operation:

$$\hat{w}_{ij} = \begin{cases} 1, & |w_{ij}| \geq 0.1 \cdot \bar{M}_{\mathrm{diff}} \\ 0, & \text{otherwise} \end{cases} \forall w_{ij} \in \boldsymbol{M}_{\mathrm{diff}} \tag{10}$$

where $|\cdot|$ denotes the absolute value operation, $\hat{w}_{ij} \in \boldsymbol{M}$, and $\bar{\boldsymbol{M}}_{\mathrm{diff}}$ represents the average value of $\boldsymbol{M}_{\mathrm{diff}}$. The nonmotion area mask $\hat{\boldsymbol{M}}$ can be obtained by subtracting M from 1. It is worth noting that the threshold is set to $0.1 \cdot \bar{\boldsymbol{M}}_{\mathrm{diff}}$ instead of 0, which can effectively avoid errors caused by satellite platform disturbances. After that, $\boldsymbol{F}^0_{\mathrm{LR}}$ and $\boldsymbol{F}^1_{\mathrm{LR}}$ are masked as follows:

$$\boldsymbol{F}^0_{\mathrm{Mask}} = \boldsymbol{F}^0_{\mathrm{LR}} \odot \boldsymbol{M} \tag{11}$$

$$\boldsymbol{F}^1_{\mathrm{Mask}} = \boldsymbol{F}^1_{\mathrm{LR}} \odot \boldsymbol{M} \tag{12}$$

where $\odot$ denotes the element-wise multiplication. Next, $\boldsymbol{F}^0_{\mathrm{Mask}}$, $\boldsymbol{F}^0_{\mathrm{LR}}$, $\boldsymbol{F}^1_{\mathrm{Mask}}$, $\boldsymbol{F}^1_{\mathrm{LR}}$, and t are fed into the temporal-aware warping module (TWM) to generate a new intermediate feature corresponding to the tth frame. To accurately predict the features of the intermediate frame, we employ deformable convolution (Deconv) to perform feature warping in two directions, $0 \rightarrow 1$ and $1 \rightarrow 0$. $0 \rightarrow 1$ is defined as the forward direction, $1 \rightarrow 0$ is defined as the backward direction. Based on the given t, $\boldsymbol{F}^0_{\mathrm{Mask}}$, and $\boldsymbol{F}^1_{\mathrm{Mask}}$, we warp $\boldsymbol{F}^0_{\mathrm{LR}}$ in the forward direction $0 \rightarrow 1$ and $\boldsymbol{F}^1_{\mathrm{LR}}$ in the backward direction $1 \rightarrow 0$ as follows:

$$\boldsymbol{F}^{0\rightarrow t}_{\mathrm{LR}} = f^{0\rightarrow1}_{\mathrm{TWM}}\left[\boldsymbol{F}^0_{\mathrm{LR}}, t, \left(\boldsymbol{F}^0_{\mathrm{Mask}}, \boldsymbol{F}^1_{\mathrm{Mask}}\right)\right] \tag{13}$$

$$\boldsymbol{F}^{1\rightarrow t}_{\mathrm{LR}} = f^{1\rightarrow0}_{\mathrm{TWM}}\left[\boldsymbol{F}^1_{\mathrm{LR}}, 1-t, \left(\boldsymbol{F}^0_{\mathrm{Mask}}, \boldsymbol{F}^1_{\mathrm{Mask}}\right)\right] \tag{14}$$

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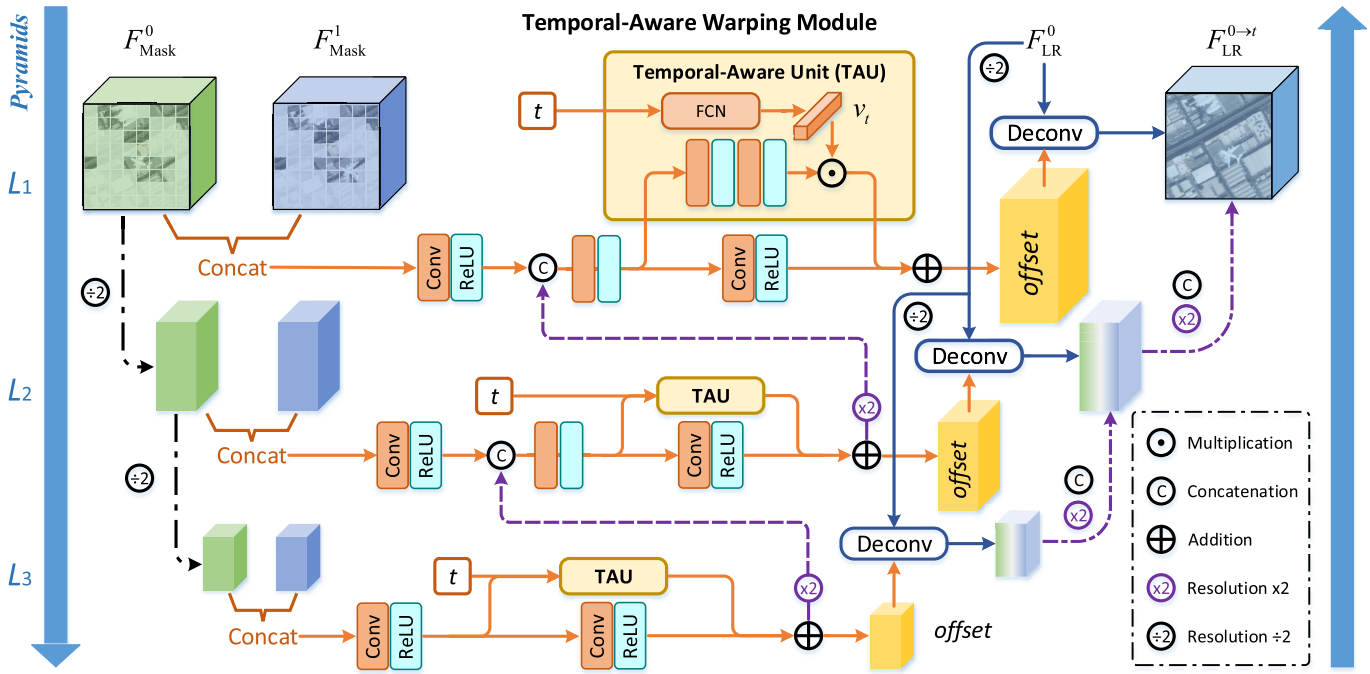


Fig. 3. Detail structure of the TWM.

where $\boldsymbol{F}_{\mathrm{LR}}^{0\to t}$ and $\boldsymbol{F}_{\mathrm{LR}}^{1\to t}$ are the interpolated features of the intermediate frame at time t, which warped from the features $\boldsymbol{F}_{\mathrm{LR}}^{0}$ and $\boldsymbol{F}_{\mathrm{LR}}^{1}$, respectively. $f_{\mathrm{TWM}}^{(\cdot)}$ denotes the operation of the TWM. The detailed structure of the TWM is illustrated in Fig. 3, where annotations "Resolution $\times 2$" and "Resolution $\div 2$" denote the operations of resolution upsampling by a factor of 2 and resolution downsampling by a factor of 2, respectively.

As shown in Fig. 3, the TWM consists of three levels (L_1–L_3), which can estimate motion and calculate offsets at different scales. First, $(\boldsymbol{F}_{\mathrm{Mask}}^{0}, \boldsymbol{F}_{\mathrm{Mask}}^{1})$ are used to calculate the motion offset between the two frames for deformable alignment. The mask guides the network to focus on motion regions, thereby improving the accuracy of offset prediction. Then, based on the temporal-aware unit (TAU), the time factor t is used to dynamically control feature warping from $0 \to 1$, while $1-t$ is used to dynamically control feature warping from $1 \to 0$. Here, taking $0 \to 1$ as an example, we illustrate how TAU dynamically controls feature warping in TWM based on the given time t. The process for $1 \to 0$ follows similarly.

As we all know, during deformable alignment, the offset plays a decisive role in determining the accuracy of feature warping. Therefore, to enable flexible warping along the temporal dimension, we embed TAU into each level of TWM to adjust the offset before performing Deconv. In TAU, we use a fully connected network (FCN) to map the time factor t to a high-dimensional vector $\boldsymbol{v}_t$. To fully exploit the motion cues for precise feature warping, we allow the offset to propagate among three levels for refinement. Subsequently, the refined offset is modulated by multiplying it with the vector $\boldsymbol{v}_t$ along the channel dimension, thereby generating a new adjusted offset. Finally, this adjusted offset is used to perform Deconv to complete feature warping.

So far, we have obtained forward warping feature $\boldsymbol{F}_{\mathrm{LR}}^{0\to t}$ and backward warping feature $\boldsymbol{F}_{\mathrm{LR}}^{1\to t}$. Finally, the interpolated feature of the intermediate frame at time t is obtained through feature fusion as follows:

$$\boldsymbol{F}_{\mathrm{LR}}^{t} = \mathtt{Conv}_{1\times 1}\left(\mathbb{C}\left[\boldsymbol{F}_{\mathrm{LR}}^{0\to t}, \boldsymbol{F}_{\mathrm{LR}}^{1\to t}\right]\right) \tag{15}$$

where $\boldsymbol{F}_{\mathrm{LR}}^{t}$ is the final interpolated features of intermediate frame at time t.

D. Interframe Deformable Attention Module

Considering that the output of MTWM lacks fine-grained details, we employ IDAM to further model interframe feature dependencies and fully exploit complementary information for refined feature representation. Specifically, our IDAM module consists of two steps. In the first step, IDA is employed to capture interframe feature dependencies and generate preliminary fusion results. In the second step, we used TEBlock to further enhance the feature representation of weak textures in preliminary fusion results, yielding the final refined features. The details of each stage are elaborated as follows.

1) IDA: The process of IDA is illustrated in Fig. 2. Specifically, two adjacent frames both need to perform IDA with the interpolated frame. Taking $\boldsymbol{F}_{\mathrm{LR}}^{0}$ and $\boldsymbol{F}_{\mathrm{LR}}^{t}$ as an example, we first locate the reference point in $\boldsymbol{F}_{\mathrm{LR}}^{0}$ and $\boldsymbol{F}_{\mathrm{LR}}^{t}$ by adaptive downsampling. The reference points are linearly spaced 2-D coordinates $\{(0,0),\ldots,(x,y)\}$. These coordinates are then normalized to the range $[-1,+1]$ based on the grid shape $x \times y$, where $(-1,-1)$ corresponds to the top-left corner and $(+1,+1)$ corresponds to the bottom-right corner. To obtain the offset of each reference point $\boldsymbol{p}$ in $\boldsymbol{F}_{\mathrm{LR}}^{0}$ to correspond to $\boldsymbol{F}_{\mathrm{LR}}^{t}$, on the one hand, we use a two-layer convolutional network to perceive the offset $\Delta\boldsymbol{p}_1$ between two frames; on the other hand, $\boldsymbol{F}_{\mathrm{LR}}^{t}$ is projected linearly to the query tokens $\boldsymbol{Q}$, and then fed into

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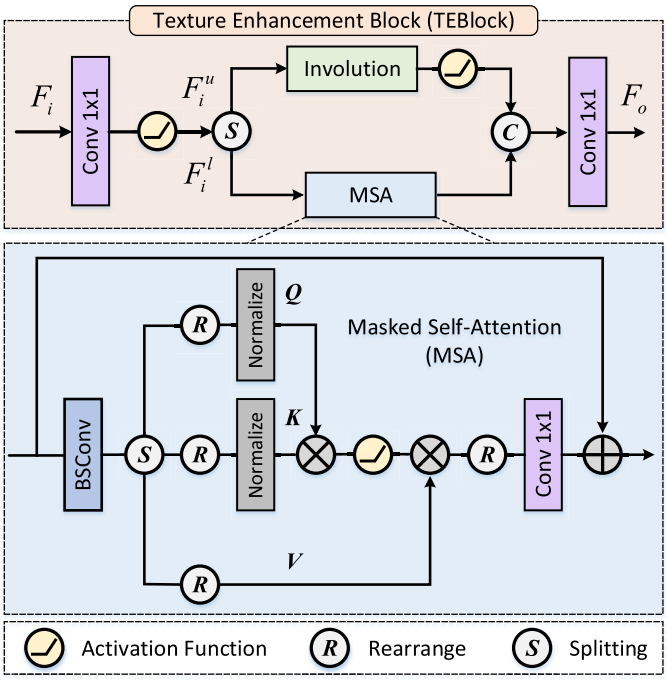


Fig. 4. Detail structure of the TEBlock.

a small convolutional network to predict the query offset $\Delta \boldsymbol{p}_2$ as follows:

$$\Delta \boldsymbol{p}_1 = \varphi_{\text{offset}}\left(\boldsymbol{F}_{\text{LR}}^0, \boldsymbol{F}_{\text{LR}}^t\right) \tag{16}$$

$$\boldsymbol{Q} = \boldsymbol{F}_{\text{LR}}^t \cdot \boldsymbol{W}_q, \Delta \boldsymbol{p}_2 = \varphi_{\text{offset}}\left(\boldsymbol{Q}\right) \tag{17}$$

where φ_{offset} is the offset prediction network, which is implemented as a lightweight convolutional subnetwork: $\texttt{Conv(9)} \rightarrow \texttt{LayerNorm} \rightarrow \texttt{GELU} \rightarrow \texttt{Conv(1)}$. Specifically, it consists of a depthwise convolution with a kernel size of 9 and a stride of 8 to aggregate local spatial context, followed by layer normalization and a GELU activation. A final 1×1 convolution is then applied to project the features into a two-channel offset map, corresponding to the horizontal and vertical displacement fields. Then, we sample the feature $\boldsymbol{F}_{\text{LR}}^0$ at the position of the deformable point to obtain feature $\tilde{\boldsymbol{F}}_{\text{LR}}^0$ and then embed it into the projection matrix

$$\tilde{\boldsymbol{F}}_{\text{LR}}^0 = \mathcal{G}\left(\boldsymbol{F}_{\text{LR}}^0; \boldsymbol{p} + \Delta \boldsymbol{p}_1 + \Delta \boldsymbol{p}_2\right) \tag{18}$$

$$\tilde{\boldsymbol{K}} = \tilde{\boldsymbol{F}}_{\text{LR}}^0 \cdot \boldsymbol{W}_k, \tilde{\boldsymbol{V}} = \tilde{\boldsymbol{F}}_{\text{LR}}^0 \cdot \boldsymbol{W}_v \tag{19}$$

where $\tilde{\boldsymbol{K}}$ and $\tilde{\boldsymbol{V}}$ are the deformed key matrix and value matrix, respectively, and $\mathcal{G}(\cdot;\cdot)$ denotes the grid sampling operation. Here, we adopt bilinear sampling for differentiability. After that, we perform multihead attention on $\boldsymbol{Q}$, $\tilde{\boldsymbol{K}}$, and $\tilde{\boldsymbol{V}}$ based on the relative position offsets $\boldsymbol{R}$ as follows:

$$\boldsymbol{Z} = \left(\delta\left(\boldsymbol{Q} \otimes \tilde{\boldsymbol{K}}^{\text{T}} / \sqrt{d} + \mathcal{G}\left(\tilde{\boldsymbol{B}}; \boldsymbol{R}\right)\right) \otimes \tilde{\boldsymbol{V}}\right) \cdot \boldsymbol{W}_o \tag{20}$$

where δ denotes the softmax function, $\otimes$ denotes the matrix multiplication, $\boldsymbol{W}_o$ is the output projection matrix, $\tilde{\boldsymbol{B}}$ is a relative position bias table. The operation of $\mathcal{G}(\tilde{\boldsymbol{B}}; \boldsymbol{R})$ follows previous work [58].

2) TEBlock: As shown in Fig. 4, our TEBlock consists of two branches, where the annotation "Activation Function" denotes the nonlinear activation applied to features, "Rearrange" indicates feature rearrangement to adapt feature layouts for subsequent operations, and "Splitting" represents channel-wise feature splitting for dual-branch processing. In the upper branch, we incorporate the involution operator [59], whose kernels are dynamically generated to capture local spatial details in satellite videos. Furthermore, the involution operator can enhance the feature representation of weak textures by increasing the kernel size to capture a wider range of local context without adding additional parameters. In the lower branch, we design a masked self-attention (MSA) to select the more useful similarity values, thereby exploiting essential global dependencies for texture reconstruction. Specifically, given an input feature $\boldsymbol{F}_{\text{in}}$, we first perform feature embedding using a 1×1 convolution, followed by activation and channel-wise splitting as follows:

$$\boldsymbol{F}_{\text{in}}^{\text{e}} = \texttt{GeLU}\left(\texttt{Conv}_{1\times1}\left(\boldsymbol{F}_{\text{in}}\right)\right) \tag{21}$$

$$\left[\boldsymbol{F}_{\text{in}}^{\text{u}}, \boldsymbol{F}_{\text{in}}^{\text{d}}\right] = \texttt{Splitting}\left(\boldsymbol{F}_{\text{in}}^{\text{e}}\right) \tag{22}$$

where $\boldsymbol{F}_{\text{in}}^{\text{u}}$ and $\boldsymbol{F}_{\text{in}}^{\text{d}}$ are the inputs of the upper and lower branches, respectively; $\texttt{GeLU}$ denotes the activation function; and $\texttt{Splitting}$ represents the channel splitting operation. In the upper branch, we introduced the involution operator [59] to enhance local feature representations. The operation is defined as follows:

$$\boldsymbol{F}_{\text{out}}^{\text{u}} = \texttt{GeLU}\left(\texttt{Involution}_{7\times7}\left(\boldsymbol{F}_{\text{in}}^{\text{u}}\right)\right) \tag{23}$$

where $\boldsymbol{F}_{\text{out}}^{\text{u}}$ denotes the output of the upper branch and $\texttt{Involution}_{7\times7}$ represents the involution operator with a kernel of 7. In the lower branch, we adopt MSA to capture global feature cues, which can fully utilize global context to model weak textures. Finally, a 1×1 convolution is employed to fuse the outputs from the two branches. The detailed process is as follows:

$$\boldsymbol{F}_{\text{out}}^{\text{d}} = \texttt{MSA}\left(\boldsymbol{F}_{\text{in}}^{\text{d}}\right) \tag{24}$$

$$\boldsymbol{F}_{\text{out}} = \texttt{Conv}_{1\times1}\left(\mathbb{C}\left[\boldsymbol{F}_{\text{out}}^{\text{u}}, \boldsymbol{F}_{\text{out}}^{\text{d}}\right]\right) \tag{25}$$

where $\boldsymbol{F}_{\text{out}}$ is the output feature and $\texttt{MSA}(\cdot)$ represents the MSA operation. The primary distinction between MSA and self-attention lies in the fact that MSA explores feature correlations over a global scope. However, due to the global scope, many positions exhibit weak or irrelevant dependencies, which hinders the extraction of meaningful feature relationships. Thus, it is necessary to apply a masking strategy on the attention map, allowing the model to adaptively select the more relevant similarity values. The process is as follows:

$$\boldsymbol{M} = \left(\boldsymbol{Q} \otimes (\boldsymbol{K})^T / \alpha\right) \tag{26}$$

$$\hat{\boldsymbol{M}} = \texttt{ReLU}\left(\boldsymbol{M}\right), \hat{\boldsymbol{V}} = \left(\hat{\boldsymbol{M}} \otimes \boldsymbol{V}\right) \tag{27}$$

where $\boldsymbol{M}$ is the attention map, α is a temperature parameter, and $(\cdot)^T$ represents the matrix transpose operation.

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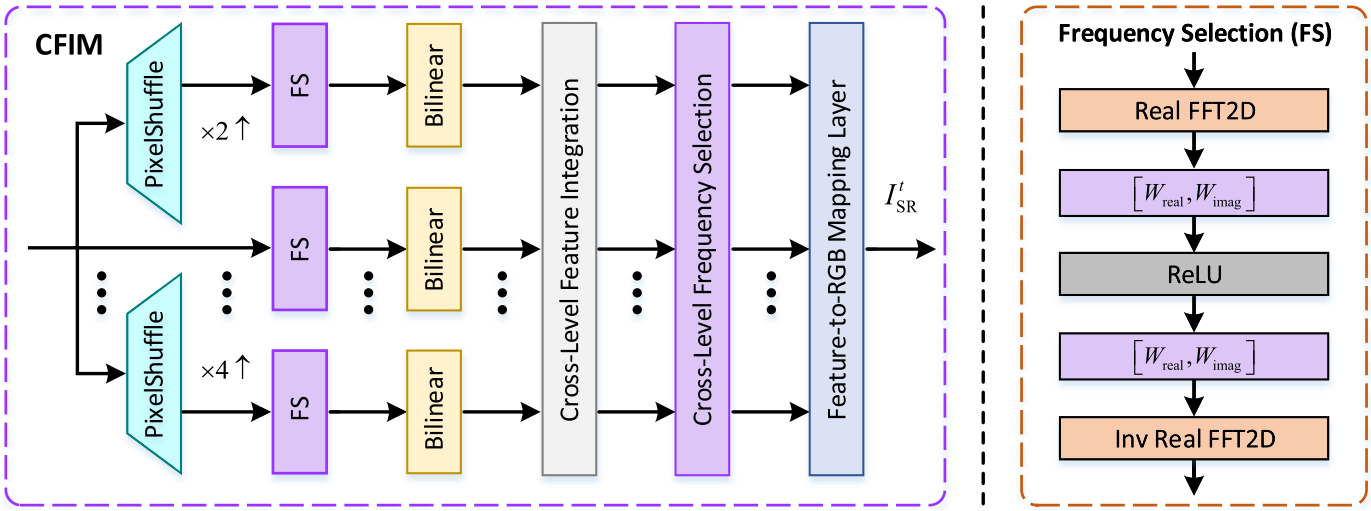


Fig. 5. Detail structure of the CFIM.

E. Cross-Level Frequency Integration Module

From a perceptual perspective, the visual world is inherently continuous, as human vision can perceive objects at arbitrary spatial location coordinates. Analogously, a video frame can be regarded as a continuous signal defined over a 2-D spatial domain, while conventional digital images only provide discrete samples on a finite pixel grid. If the pixel value (or feature representation) at an arbitrary spatial coordinate can be inferred, a continuous representation of the video frame in the spatial dimension can be obtained. However, although such a coordinate-based formulation enables continuous spatial modeling, reconstructing high-quality video frames at arbitrary resolutions still critically depends on the availability of sufficient high-frequency information. As we all know, video frames at different resolutions exhibit distinct spectral characteristics after FFT, where HR frames contain richer high-frequency information. Inspired by this work [60], we propose the CFIM to adaptively integrate high-frequency components from cross-level resolution feature maps through FS, effectively learn the high-frequency information required for reconstructing video frames at arbitrary coordinates, thereby achieving continuous-scale SR reconstruction.

Specifically, as shown in Fig. 5, we first use pixelshuffle [61] to perform upsampling and generate a group of cross-level feature maps $\{\boldsymbol{F}^n_{\text{map}} \in \mathbb{R}^{C\times H(2^{n-1})\times W(2^{n-1})}\}^N_{n=1}$. Each level of $\{\boldsymbol{F}^n_{\text{map}} \in \mathbb{R}^{C\times H(2^{n-1})\times W(2^{n-1})}\}^N_{n=1}$ has the different resolution, where H and W represent the height and width of the LR video frame, respectively. Subsequently, at each level, we apply the FS mechanism to enhance the salient features corresponding to this level as follows:

$$\left\{\boldsymbol{F}^n_{\text{FS}}\right\}^N_{n=1} = \mathcal{FS}\left(\left\{\boldsymbol{F}^n_{\text{map}}\right\}^N_{n=1}\right) \tag{28}$$

where $\mathcal{FS}(\cdot)$ denotes the FS operation. This process is crucial, as it selectively isolates and amplifies the frequency components that most contribute to the textural information unique to that level, thereby significantly improving its feature representation power. Finally, we perform cross-level FS integration to complete the mapping from multiresolution space to continuous resolution space. The detailed process is as follows.

Given an arbitrary upsampling scale factor r, we first establish the correspondence between the LR coordinate system and the target SR coordinate system and calculate the corresponding 2-D coordinates $\mathcal{C}_r$ of SR results. Different upsampling scale factors r correspond to different coordinate distributions in the SR space. As r increases continuously, the sampling density of pixel coordinates in the SR space also increases in a continuous manner. Then, for an arbitrary coordinate $c \in \mathcal{C}_r$, as long as we get its RGB value, we have the final SR result. To achieve this, we propose to use a set of cross-level feature maps $\{\boldsymbol{F}^n_{\text{map}} \in \mathbb{R}^{C\times H(2^{n-1})\times W(2^{n-1})}\}^N_{n=1}$ along with c to calculate its corresponding RGB value. Given an arbitrary query coordinate c, it has the nearest four latent codes at each level, defined as $\{\boldsymbol{O}^n_i\}^4_{i=1}$ and $\{\boldsymbol{O}^n_i \in \boldsymbol{F}^n_{\text{FS}}\}^N_{n=1}$. At each level, we interpolate these four latent codes through the bilinear interpolation function $\mathcal{B}$ to generate the feature representation at coordinate c

$$\boldsymbol{O}^n_c = \mathcal{B}\left(\left\{\boldsymbol{O}^n_i\right\}^4_{i=1}\right) \quad \forall n \in N, \tag{29}$$

$$\mathcal{B}\left(\left\{\boldsymbol{O}^n_i\right\}^4_{i=1}\right) := [1-d_1, d_1]\begin{bmatrix}\boldsymbol{O}^n_1 & \boldsymbol{O}^n_2\\ \boldsymbol{O}^n_3 & \boldsymbol{O}^n_4\end{bmatrix}\begin{bmatrix}1-d_2\\ d_2\end{bmatrix} \tag{30}$$

where d_1 and d_2 denote the normalized horizontal and vertical distance of c relative to $\boldsymbol{O}^n_1$. Next, we concatenate $\{\boldsymbol{O}^n_c\}^N_{n=1}$ along the channel dimension. After that, FS is applied to adaptively choose the appropriate cross-level latent codes as follows:

$$\boldsymbol{O}_c = \mathbb{C}\left[\boldsymbol{O}^1_c, \boldsymbol{O}^2_c, \ldots, \boldsymbol{O}^N_c\right] \tag{31}$$

$$\boldsymbol{O}_{\text{F}} = \texttt{FFT}\left(\boldsymbol{O}_c\right) \tag{32}$$

$$\hat{\boldsymbol{O}}_{\text{F}} = \boldsymbol{O}_{\text{F}} \otimes \left(\boldsymbol{W}^1_{\text{real}} + i \cdot \boldsymbol{W}^1_{\text{imag}}\right) \tag{33}$$

$$\hat{\boldsymbol{O}}_{\text{c}} = \texttt{IFFT}\left(\texttt{ReLU}\left(\hat{\boldsymbol{O}}_{\text{F}}\right) \otimes \left(\boldsymbol{W}^2_{\text{real}} + i \cdot \boldsymbol{W}^2_{\text{imag}}\right)\right) \tag{34}$$

where $\boldsymbol{W}^1_{\text{real}}, \boldsymbol{W}^1_{\text{imag}}, \boldsymbol{W}^2_{\text{real}}$, and $\boldsymbol{W}^2_{\text{imag}}$ are the predefined FS weights. For every $c \in \mathcal{C}_r$, FS is performed to obtain feature representations $\hat{\boldsymbol{O}}_{\mathcal{C}_r}$ corresponding to all coordinates under the

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upsampling scale r. Finally, we use two convolutional layers to map the refined features to pixel values as follows:

$$\boldsymbol{I}_{\mathrm{SR}}^{t} = \mathtt{Conv}_{3\times 3}\left(\mathtt{LeakyReLU}\left(\mathtt{Conv}_{1\times 1}\left(\hat{\boldsymbol{O}}_{\boldsymbol{C}_r}\right)\right)\right). \tag{35}$$

F. Loss Function

To stabilize the training process, following previous work, we use the Charbonnier penalty function [63] to optimize our network. Moreover, to ensure the accuracy of temporal-aware feature warping, an additional LR supervision term is incorporated as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{SR}} + \alpha \cdot \mathcal{L}_{\mathrm{LR}} \tag{36}$$

where $\mathcal{L}_{\mathrm{SR}}$ denotes the SR reconstruction loss, $\mathcal{L}_{\mathrm{LR}}$ represents the additional LR supervision loss, and α is the hyperparameter to balance the two subitems.

1) SR Reconstruction Loss: To ensure the generalization performance of the network across various scales, we uniformly sample scaling factors within the range $[2, 4]$, thereby providing sufficient supervision during training as follows:

$$\mathcal{L}_{\mathrm{SR}} = \mathbb{E}_{r\sim\mathcal{U}(2,4)}^{t}\left(\sqrt{\left(\boldsymbol{I}_{\mathrm{SR}}^{t} - \boldsymbol{I}_{\mathrm{HR}}^{t}\right)^{2} + \varepsilon^{2}}\right) \tag{37}$$

where $\mathcal{U}(2, 4)$ denotes a uniform distribution over $[2, 4]$ and $\mathbb{E}$ represents the expectation (mean) operator.

2) LR Supervision Loss: To ensure the accuracy of temporal-aware feature warping, we propose mapping the warped features back to the LR video frames and applying supervision using the corresponding video frame at time t as follows:

$$\mathcal{L}_{\mathrm{LR}} = \mathbb{E}^{t}\left(\sqrt{\left(f_{\theta}\left(\boldsymbol{F}_{\mathrm{LR}}^{t}\right) - \boldsymbol{I}_{\mathrm{LR}}^{t}\right)^{2} + \varepsilon^{2}}\right) \tag{38}$$

where f_{θ} denotes the function that maps features $\boldsymbol{F}_{\mathrm{LR}}^{t}$ to pixel values and $\mathbb{E}$ represents the expectation (mean) operator.

IV. Experimental Results

In this section, we first introduce the datasets, evaluation metrics, and training details. Then, we present a comprehensive comparison between our proposed method and several state-of-the-art (SOTA) methods.

A. Datasets and Metrics

Our video data come from the Jilin-1, Carbonite-2, and SkySat-1 satellites, comprising 11 distinct video scenes with a spatial resolution of approximately 1 m. Our dataset is cropped from the 11 video scenes. Specifically, each video scene is cropped into clips of 640×640 resolution, yielding a total of 205 video clips, of which 179 are used for training, 10 for validation, and the remaining 16 for testing, named Clip001–Clip016. In addition, to objectively demonstrate the performance of our method, we introduce two evaluation metrics: peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) [64].

B. Implementation Details

For shallow feature extraction, the number of SFJMBs k is set to 5, while for deep feature extraction, the number of SFJMBs K is set to 10. The number of cross-level feature maps N is set to 3. During training, we employ data augmentation through random horizontal flips and rotation (90°, 180°, and 270°). α is set to 0.1. LR video clips of size 32×32 are randomly cropped as input, with a batch size of 8. Our model is trained in two stages by using the Adam optimizer [65], with a total of 200k iterations. For each video, we take nine consecutive frames to form a training sample. Specifically, in the first stage, the downsampling scale factor is fixed at $\times 4$ for 150k iterations, and the learning rate is initialized to 4×10^{-4} and decayed to 1×10^{-6} using cosine annealing every 50k iterations. In the second stage, the scale factor is randomly sampled from the uniform distribution $[2, 4]$ for 50k iterations, with the learning rate initialized to 1×10^{-5} and decayed to 1×10^{-6} via cosine annealing. The LR video frames are downsampled by the bicubic kernel. Our proposed model is implemented based on the PyTorch framework and trained on an NVIDIA GeForce RTX 3090 GPU.

C. Comparison With the SOTA VSR Methods

To evaluate the effectiveness of SV-CSTSR, we compare it with several SOTA VSR models, including TMNet (CVPR21) [48], MSTT (JAG22) [21], VideoINR (CVPR22) [52], MoTIF (CVPR23) [53], BFSTVSR (CVPR25) [62], and STNO (TIP25) [55].

For a fair comparison on the satellite video FSSTSR task, we additionally trained a model with a fixed space scale factor r, referred to as SV-CSTSR-$fixed$. In Table I, the space scale factor defaults to 4. On Clip001–Clip005, the time scale factor is fixed at 2 to test single-frame interpolation (SFI), denoted as SFI-(4, 2). On Clip006-*Center*–Clip015-*Center*, the time scale factor is set to 8 to test multiframe interpolation (MFI), denoted as MFI-(4, 8). Note that for the MFI task, we only report the PSNR and SSIM of the important intermediate interpolated frames (1st, 5th, and 9th), which are marked as $-$*Center*. From Table I, we can find that among methods of the same type (i.e., TMNet, MSTT, and SV-CSTSR-$fixed$; or VideoINR, MoTIF, BFSTVSR, and SV-CSTSR), our proposed method consistently achieves the best performance, as indicated in bold. Even when compared to the spatially fixed method TMNet, our continuous model SV-CSTSR remains highly competitive, particularly on the MFI task. Furthermore, the network parameters of our method are less than half of the comparison methods such as TMNet, VideoINR, BFSTVSR, and MoTIF, which demonstrate its superior efficiency.

We also performed a visual comparison on the satellite video FSSTSR task. As shown in Fig. 6, on the SFI-(4, 2) task, our method yields higher quality SR reconstruction results, particularly in recovering fine structural details and maintaining sharper object boundaries. Unlike competing methods, which often produce incorrect texture orientation or blurred edges, our method preserves the geometric integrity of targets such as building textures and aircraft outlines. This advantage is further reflected in the higher PSNR and SSIM metrics,

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TABLE I

PERFORMANCE COMPARISON (PSNR/SSIM) ON THE SATELLITE VIDEO FSSTSR TASK. RED AND BLUE INDICATE THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY. BOLD REPRESENTS THE BEST AMONG SIMILAR METHODS. * DENOTES OUR PROPOSED METHOD. THE SYMBOL -/- DENOTES THAT THE METHOD IS NOT APPLICABLE

Video Test Clips		Training With Fixed Scale r			Training With Continuous Scale r				
		TMNet [48]	MSTT [21]	SV-CSTSR-$fixed$*	VideoINR [52]	MoTIF [53]	BFSTVSR [62]	STNO [55]	SV-CSTSR*
#Params		12.3M	6.8M	5.6M	11.3M	12.6M	13.5M	9.4M	5.6M
Clip001	SFI-$(4, 2)$	42.48/0.9581	42.50/0.9578	**42.69/0.9597**	41.74/0.9574	41.88/0.9547	41.85/0.9548	42.17/0.9558	**42.55/0.9590**
Clip002		34.38/0.9367	34.37/0.9363	**34.68/0.9391**	33.58/0.9314	33.61/0.9298	33.70/0.9316	34.02/0.9338	**34.48/0.9379**
Clip003		31.95/0.9302	32.08/0.9309	**32.41/0.9350**	30.99/0.9235	30.91/0.9192	31.38/0.9249	31.58/0.9274	**32.17/0.9328**
Clip004		32.09/0.9202	31.79/0.9129	**32.38/0.9237**	30.66/0.9077	30.83/0.9043	31.00/0.9068	31.25/0.9094	**32.07/0.9191**
Clip005		33.94/0.8971	33.72/0.8929	**34.10/0.9006**	33.46/0.8918	33.43/0.8895	33.34/0.8916	33.74/0.8946	**33.97/0.8989**
Clip006-$Center$	MFI-$(4, 8)$	33.65/0.9430	-/-	**33.99/0.9461**	32.39/0.9352	32.60/0.9345	32.81/0.9370	32.99/0.9386	**33.71/0.9440**
Clip007-$Center$		33.89/0.9357	-/-	**34.19/0.9390**	31.96/0.9229	32.53/0.9235	32.84/0.9274	33.05/0.9295	**33.92/0.9365**
Clip008-$Center$		41.70/0.9633	-/-	**42.11/0.9665**	40.82/0.9637	40.65/0.9618	40.82/0.9615	40.85/0.9639	**41.88/0.9657**
Clip009-$Center$		33.28/0.9316	-/-	**33.51/0.9340**	32.80/0.9287	32.63/0.9255	32.63/0.9268	32.87/0.9281	**33.31/0.9323**
Clip010-$Center$		33.62/0.9330	-/-	**34.04/0.9360**	33.23/0.9305	32.78/0.9253	32.98/0.9276	33.33/0.9303	**33.77/0.9337**
Clip011-$Center$		34.54/0.9403	-/-	**35.14/0.9439**	33.72/0.9351	33.59/0.9319	33.74/0.9334	33.91/0.9351	**34.69/0.9405**
Clip012-$Center$		35.75/0.9410	-/-	**36.15/0.9440**	34.70/0.9355	34.91/0.9353	35.00/0.9370	35.26/0.9387	**35.82/0.9420**
Clip013-$Center$		34.19/0.9445	-/-	**34.57/0.9471**	33.25/0.9381	33.29/0.9363	33.21/0.9369	33.43/0.9388	**34.37/0.9450**
Clip014-$Center$		36.94/0.9467	-/-	**37.41/0.9488**	36.33/0.9428	36.12/0.9402	35.89/0.9407	36.34/0.9418	**37.06/0.9470**
Clip015-$Center$		32.65/0.9180	-/-	**32.85/0.9208**	31.95/0.9114	31.84/0.9083	31.80/0.9098	32.20/0.9129	**32.71/0.9185**

TABLE II

PERFORMANCE COMPARISON (PSNR/SSIM) ON THE SATELLITE VIDEO CSTSR TASK (ON DATASET CLIP016-*Average*). BOLD INDICATES THE BEST PERFORMANCE

Time Scale	Space Scale	VideoINR [52]	MoTIF [53]	BFSTVSR [62]	STNO [55]	SV-CSTSR (Ours)
$\times 3$	$\times 4$	32.03/0.9303	31.97/0.9270	31.93/0.9278	32.31/0.9308	**33.21/0.9384**
$\times 3$	$\times 6$	28.40/0.8471	28.13/0.8393	28.08/0.8386	28.40/0.8446	**28.72/0.8517**
$\times 3$	$\times 8$	26.47/0.7884	26.31/0.7812	26.15/0.7726	26.53/0.7869	**26.68/0.7928**
$\times 4$	$\times 4$	31.86/0.9298	31.75/0.9263	31.70/0.9272	32.17/0.9305	**33.11/0.9381**
$\times 4$	$\times 6$	28.34/0.8468	28.06/0.8388	28.01/0.8381	28.35/0.8444	**28.71/0.8517**
$\times 4$	$\times 8$	26.44/0.7883	26.29/0.7810	26.12/0.7723	26.51/0.7868	**26.68/0.7928**
$\times 5$	$\times 4$	31.71/0.9293	31.56/0.9258	31.48/0.9266	32.06/0.9300	**33.01/0.9377**
$\times 5$	$\times 6$	28.29/0.8464	27.98/0.8384	27.92/0.8376	28.28/0.8439	**28.67/0.8513**
$\times 5$	$\times 8$	26.41/0.7881	26.25/0.7807	26.08/0.7720	26.48/0.7866	**26.65/0.7925**

indicating both improved fidelity to the ground truth and enhanced perceptual consistency. As shown in Fig. 7, on the MFI-(4, 8) task, we find that our method excels in reconstructing moving targets, where other comparison methods often struggle with motion artifact and structural distortion. Specifically, our method reconstructs sharper contours and more faithful shapes of dynamic objects, such as airplanes, effectively preserving their fine-scale details even under significant motion. This improvement is primarily attributed to the proposed MTWM and IDAM, which accurately capture spatiotemporal correlations to perform effective feature warping and fusion, thereby ensuring the consistent recovery of spatial structures and maintaining temporal coherence. Consequently, our method achieves a more natural and accurate reconstruction of moving objects, significantly improving visual clarity and fidelity compared to existing baselines.

For a comprehensive comparison on the satellite video CSTSR task. Table II presents results across nine different upsampling scenarios. Here, we report the PSNR and SSIM of all intermediate interpolated frames, with Clip016 denoted as Clip016-*Average*. As shown in Table II, our method consistently outperforms existing methods across varying time and space scale factors. Notably, SV-CSTSR achieves the highest PSNR and SSIM under all experimental settings, indicating superior fidelity to ground truth and stronger perceptual quality. Moreover, as the reconstruction scale increases, the

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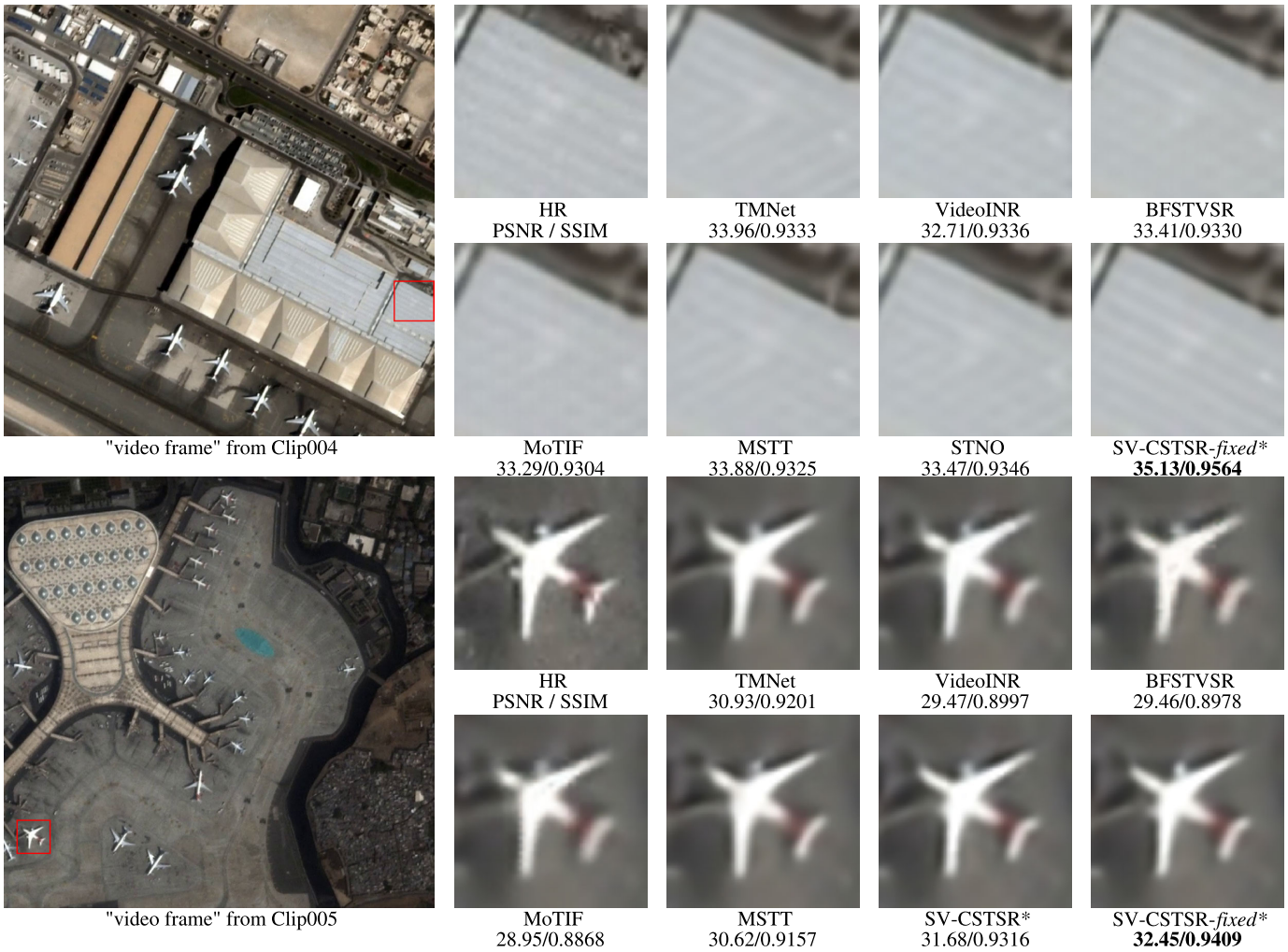


Fig. 6. Visual comparison with time scale factor 2 and space scale factor 4 on Clip004 and Clip005. More details are in the red box. * denotes our proposed method.

competitive advantage of our method becomes even more pronounced.

We also conduct a visual comparison with a time scale factor $\times 5$ and a space scale factor $\times 4$. As shown in Fig. 8, we can observe that the results generated by SV-CSTSR contain more accurate and fine-grained textures, while other methods generate excessive noise, blurred edges, and motion artifacts. These visual results further demonstrate the effectiveness of the proposed MTWM, IDAM, and $\mathcal{L}_{\mathrm{LR}}$, which collectively ensure precise feature warping and facilitate high-quality VFI.

To evaluate the performance of SV-CSTSR on noninteger space scale factors, we conducted comparative experiments on test clips Clip001 and Clip002. As shown in Table III, we can find that SV-CSTSR achieves the highest PSNR and SSIM values across nearly all experimental settings. For instance, on Clip001, SV-CSTSR consistently achieves the highest scores, surpassing others by up to nearly 1.0 dB at 2.2 scale. In Clip002, our method delivers notable gains across most scaling factors, with margins exceeding 1.2 dB at certain points.

To comprehensively evaluate the computational efficiency of the proposed SV-CSTSR, we compare its model complexity and runtime with all comparison methods on the Clip001 of task SFI-$(4, 2)$. The results were tested on an NVIDIA GeForce 3090 GPU. As reported in Table IV and illustrated in Fig. 9, SV-CSTSR achieves the lowest computational cost, requiring only 417.6 GFLOPs and 5.6M parameters, which is substantially lower than both conventional STSR methods (e.g., TMNet and MSTT) and recent CSTSR approaches such as VideoINR, MoTIF, and BFSTVSR. In addition, SV-CSTSR attains a runtime of 0.2534 s, making it the fastest method among all CSTSR models and highly competitive even compared with STSR methods. Importantly, despite its significantly reduced complexity, SV-CSTSR maintains competitive reconstruction performance, demonstrating a favorable tradeoff between accuracy and efficiency.

Finally, we conducted experiments in the challenging scenarios to validate the robustness of the proposed method. Fig. 10 presents qualitative results on two challenging scenarios: a texture-weak scene with dense and complex vehicle motion (Clip010) and a large-scale motion scene with significant interframe displacement (Clip016). In Clip010, the presence of weak textures and dense vehicle movements makes accurate motion modeling particularly challenging; nevertheless, our SV-CSTSR preserves correct structural details

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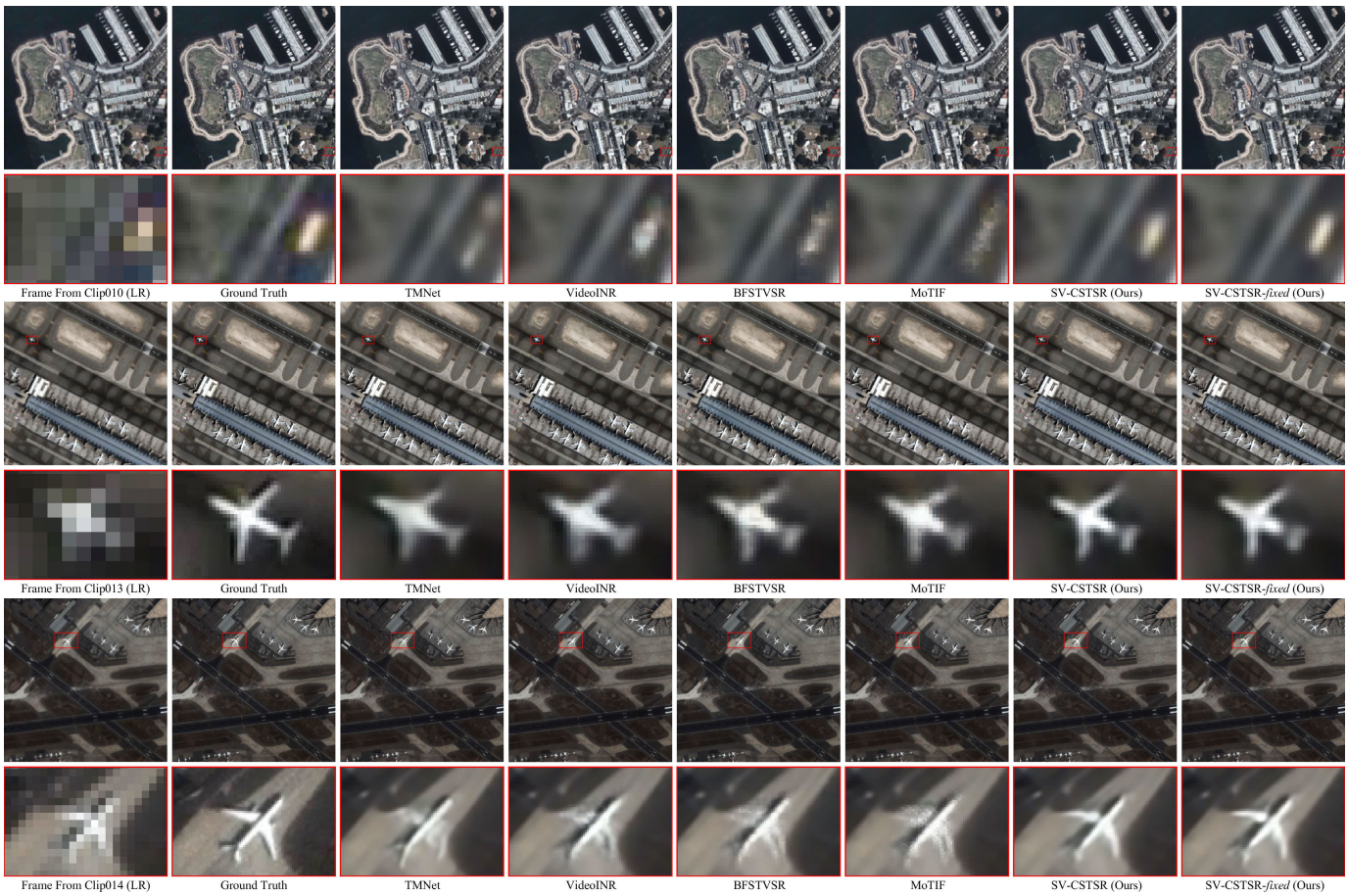


Fig. 7. Visual comparison with time scale factor 8 and space scale factor 4 on Clip010, Clip013, and Clip014. More details are in the red box.

without missing moving vehicles, indicating strong robustness to dense and complex motions. In Clip016, where the rapidly taxiing aircraft undergoes large interframe displacement, our method maintains clear object boundaries with minimal distortion. These results demonstrate the robustness of SV-CSTSR under both dense motion and large-scale motion scenarios.

V. Discussion

In this section, we first discuss the effectiveness of the loss function $\mathcal{L}_{\mathrm{LR}}$. Next, we discuss the effectiveness of the FM in SFJMB. Third, we study the effect of the mask strategy in MTWM. Subsequently, we discuss the effectiveness of the IDA and TEBlock in IDAM. Finally, we discuss the effectiveness of the FS in CFIM and investigate the settings of N.

A. Study of the LR Supervision Loss $\mathcal{L}_{\mathrm{LR}}$

To demonstrate the effectiveness of the LR supervision loss $\mathcal{L}_{\mathrm{LR}}$, we propose to remove the LR supervision loss $\mathcal{L}_{\mathrm{LR}}$ and train our network SV-CSTSR using only SR reconstruction loss $\mathcal{L}_{\mathrm{SR}}$, denoted as SV-CSTSR w/o $\mathcal{L}_{\mathrm{LR}}$. We conduct ablation studies on the SFI-$(4, 2)$ and MFI-$(4, 8)$ tasks. As shown in Table V, on the SFI-$(4, 2)$ task, SV-CSTSR w/o $\mathcal{L}_{\mathrm{LR}}$ consistently underperforms SV-CSTSR across all test clips. Furthermore, as shown in Fig. 11, SV-CSTSR continues to outperform SV-CSTSR w/o $\mathcal{L}_{\mathrm{LR}}$ on the MFI-$(4, 8)$ task. Overall, these results clearly demonstrate that the superior performance of SV-CSTSR on both SFI-$(4, 2)$ and MFI-$(4, 8)$ tasks validates the effectiveness of the LR supervision loss $\mathcal{L}_{\mathrm{LR}}$.

B. Study of the FM in SFJMB

To verify the effectiveness of the FM in SFJMB, we propose to remove the FM component while retaining the convolution operation, denoted as SV-CSTSR w/o FM. We perform ablation learning on both SFI-$(4, 2)$ and MFI-$(4, 8)$ tasks. On the SFI-$(4, 2)$ task, we report the average PSNR and SSIM over the five test clips, Clip001–Clip005. On the MFI-$(4, 8)$ task, we compare the average PSNR and SSIM across the ten test clips Clip006-*Center*–Clip015-*Center*. As shown in Table VI, we can observe that our FM can significantly improve network performance, owing to its strong capability of modeling global dependencies in frequency domains.

C. Study of the Mask Strategy in MTWM

To demonstrate the effectiveness of the mask strategy in MTWM, we conduct ablation experiments under varying time scale factors. Since the mask strategy is designed to locate motion regions, it enables more accurate offset estimation

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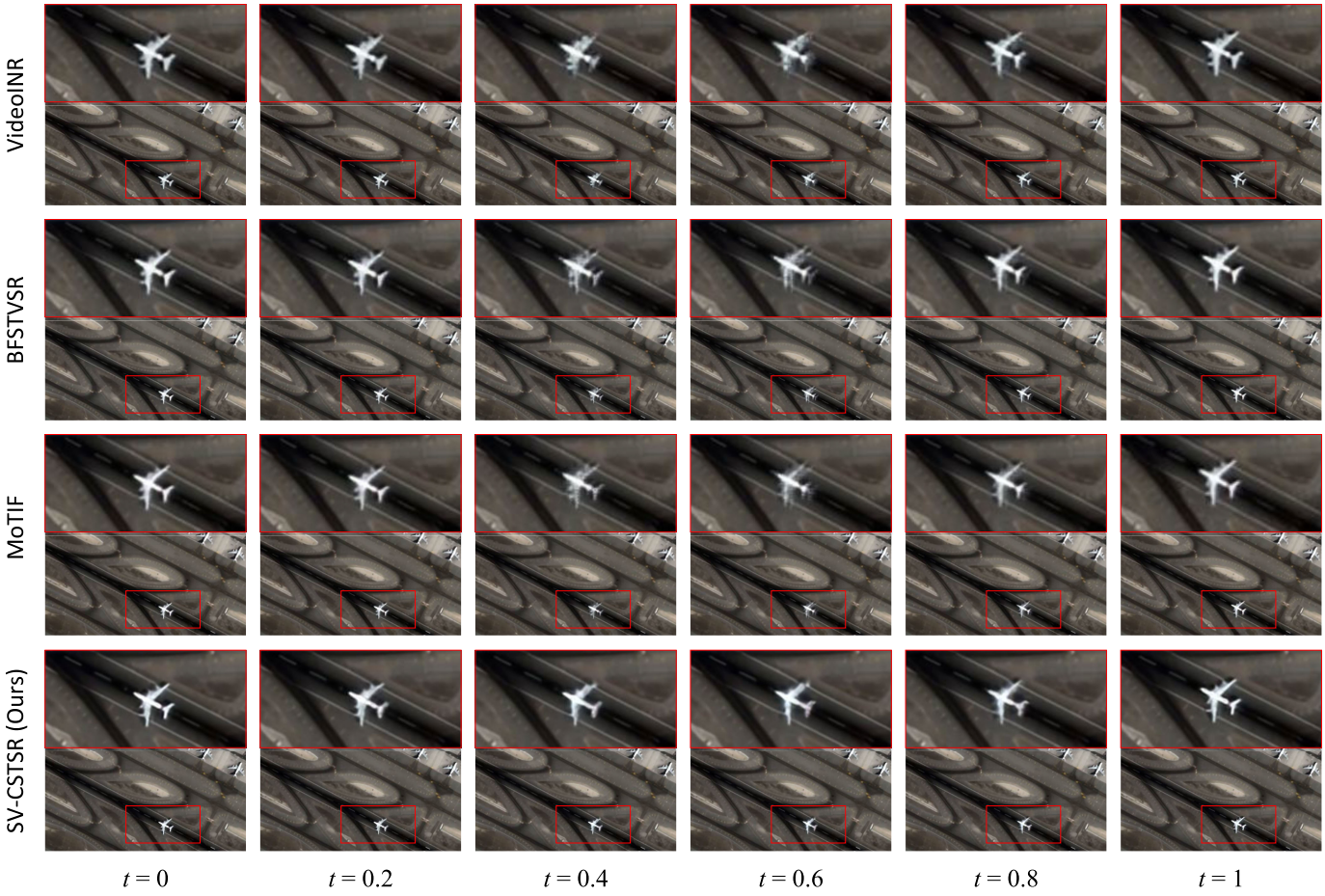


Fig. 8. Visual comparison with time scale factor 5 and space scale factor 4 on Clip016. The patches for comparison are marked with red boxes.

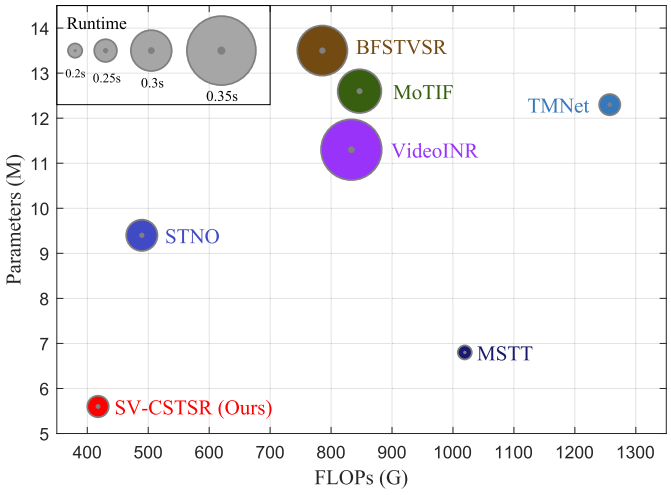


Fig. 9. Comparison of network parameters, floating-point operations (FLOPs), and runtime on Clip001 of Task SFI-(4, 2). VideoINR, MoTIF, BFSTVSR, STNO, and SV-CSTSR are CSTSR methods, while TMNet and MSTT are STSR methods because they cannot achieve CSTSR.

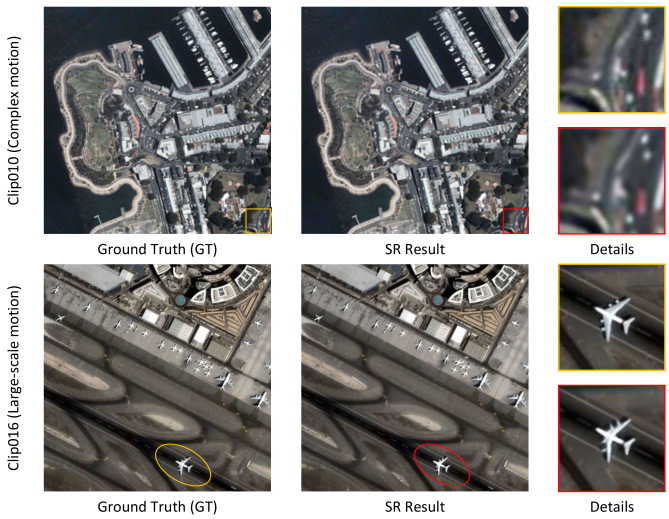


Fig. 10. Validation of the challenging scenarios: complex motion and large-scale motion on the task SFI-(4, 2). Clip010 contains dense vehicle motion and has weak textures, while Clip016 includes an aircraft rapidly taxiing on the runway.

and improves the precision of feature warping. Therefore, comparing interpolation results under different time scale factors serves as the most direct means of validating its effectiveness. Specifically, we fix the space scale factor to 4, while setting the time scale factors to 3, 4, and 5, corresponding to MFI-(4, 3), MFI-(4, 4), and MFI-(4, 5), respectively. The experimental results are shown in Table VII, SV-CSTSR consistently outperforms its counterpart without the mask strategy

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TABLE III

PERFORMANCE COMPARISON (PSNR/SSIM) ON TWO SATELLITE VIDEO TEST CLIPS (CLIP001 AND CLIP002) WITH $t = 2$ AND CONTINUOUS SPACE SCALES r. RED AND BLUE INDICATE THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY. * DENOTES OUR PROPOSED METHOD

Clips	r	VideoINR [52]	MoTIF [53]	BFSTVSR [62]	SV-CSTSR*
Clip001	2.2	43.51/0.9770	43.99/0.9761	44.24/0.9762	44.98/0.9780
	2.4	43.40/0.9762	43.76/0.9749	44.01/0.9752	44.72/0.9770
	2.6	43.26/0.9745	44.16/0.9749	43.63/0.9739	44.25/0.9757
	2.8	43.08/0.9731	43.89/0.9733	43.40/0.9723	43.98/0.9742
	3.0	43.47/0.9728	43.97/0.9720	43.77/0.9716	44.53/0.9738
	3.1	42.67/0.9700	43.39/0.9699	42.95/0.9691	43.48/0.9711
	3.3	42.44/0.9681	42.46/0.9659	42.66/0.9664	43.33/0.9689
	3.5	42.60/0.9659	42.82/0.9645	42.75/0.9642	43.50/0.9671
	3.7	41.82/0.9618	42.34/0.9609	41.99/0.9603	42.51/0.9630
	3.9	41.56/0.9588	41.45/0.9556	41.64/0.9561	42.34/0.9601
Clip002	2.1	35.23/0.9651	36.68/0.9704	35.89/0.9674	36.42/0.9700
	2.3	36.75/0.9704	38.03/0.9724	37.91/0.9723	38.97/0.9750
	2.5	37.40/0.9709	39.07/0.9733	38.46/0.9722	39.77/0.9752
	2.7	36.88/0.9677	37.91/0.9697	37.31/0.9680	38.02/0.9707
	2.9	36.76/0.9656	37.78/0.9671	37.44/0.9664	38.46/0.9696
	3.2	36.02/0.9592	36.67/0.9600	36.48/0.9598	37.41/0.9635
	3.4	35.44/0.9536	35.71/0.9535	35.42/0.9526	36.03/0.9565
	3.6	33.00/0.9381	33.34/0.9381	33.55/0.9400	34.42/0.9462
	3.8	33.74/0.9373	33.75/0.9360	33.45/0.9350	33.80/0.9394
	4.0	33.58/0.9314	33.61/0.9298	33.70/0.9316	34.48/0.9379

TABLE IV

COMPARISON OF FLOPS AND RUNTIME ON CLIP001 OF TASK SFI-(4, 2)

Methods		Runtime (s)	FLOPs (G)
STSR	TMNet [48]	0.2375	1257.2
	MSTT [21]	0.1603	1019.4
CSTSR	VideoINR [52]	0.3378	833.3
	MoTIF [53]	0.3065	846.7
	BFSTVSR [62]	0.3195	785.7
	STNO [55]	0.2746	489.5
	SV-CSTSR (Ours)	0.2534	417.6

across all tasks. Specifically, incorporating the mask strategy yields a 0.21-dB improvement in PSNR, with performance gains becoming increasingly significant as the time scale factor grows. These results clearly demonstrate that the mask strategy plays a crucial role in guiding the network to concentrate on motion regions, thereby enhancing reconstruction quality.

D. Study of the IDA and TEBlock in IDAM

To demonstrate the effectiveness of the IDA and TEBlock in IDAM, ablation experiments are carried out. Specifically, we first remove the IDA to construct a variant termed Model-I and then remove the TEBlock to obtain another variant termed Model-II. The experimental results, presented in Table VIII, reveal that both IDA and TEBlock contribute positively to the network reconstruction performance. Moreover, their combination within SV-CSTSR achieves the best results, highlighting their strong complementarity and synergistic effect.

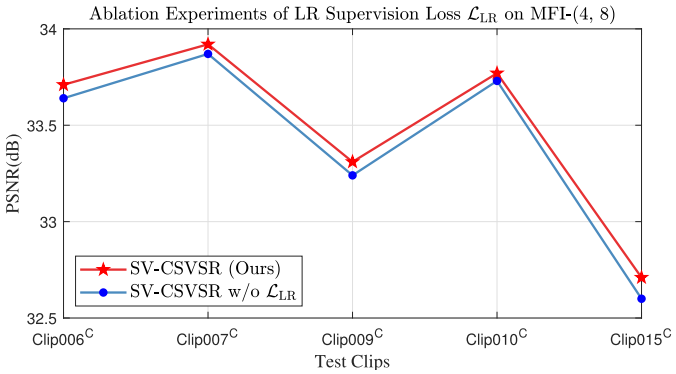


Fig. 11. Ablation study of $\mathcal{L}_{LR}$ on Clip006-*Center*–Clip015-*Center* (only five test clips with similar PSNR values were selected for plotting). Clip006^C denotes Clip006-*Center*. Our experiments were performed on the MFI-(4, 8) task.

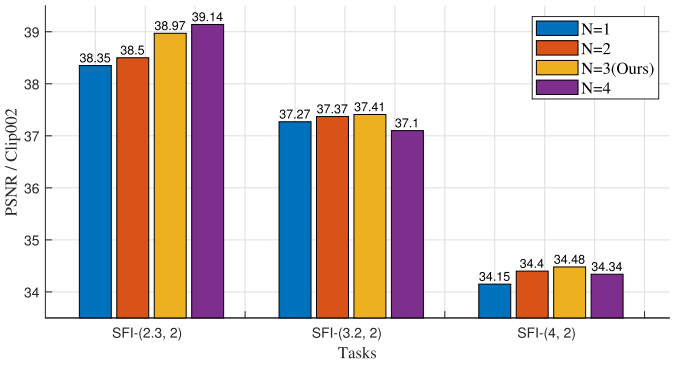


Fig. 12. Ablation experiments of the value N of CFIM on clip002 in terms of PSNR.

E. Study of the FS and Settings of N in CFIM

We conduct an ablation experiment to validate the rationality and superiority of FS mechanism in CFIM. Since our CFIM is used for upsampling, we conduct comprehensive verification on multiple different space scale factors. Specifically, in this ablation experiment, we fix the time scale factor to 4, and set the space scale factors to 4, 6, and 8, marked as MFI-(4, 4), MFI-(6, 4), and MFI-(8, 4), respectively. From Table IX, we can conclude that our FS can effectively improve network reconstruction performance across multiple space scale factors. Furthermore, the experimental results also prove that FS can more effectively use latent codes to learn continuous feature representations.

In addition, we further investigate the impact of the parameter N in CFIM. Specifically, we fix the time scale factor to 2 and set the space scale factors to 2.3, 3.2, and 4, marked as SFI-(2.3,2), SFI-(3.2,2), and SFI-(4,2), respectively. As illustrated in Figs. 12 and 13, the reconstruction performance improves with increasing N when $N < 3$. This improvement

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TABLE V

ABLATION STUDY OF $\mathcal{L}_{\mathrm{LR}}$ ON CLIP001–CLIP005. OUR EXPERIMENTS WERE PERFORMED ON THE SFI-(4, 2) TASK. BOLD INDICATES THE BETTER RESULTS

Models	Clip001	Clip002	Clip003	Clip004	Clip005
SV-CSTSR w/o $\mathcal{L}_{\mathrm{LR}}$	42.42/0.9587	34.46/0.9370	32.08/0.9312	32.03/0.9180	33.92/0.8971
SV-CSTSR	**42.55/0.9590**	**34.48/0.9379**	**32.17/0.9328**	**32.07/0.9191**	**33.97/0.8989**

TABLE VI

ABLATION STUDY OF FM IN SFJMB. AVE-DENOTES AVERAGE VALUE

Models Tasks	Ave-Clip(001-005) SFI-$(4, 2)$	Ave-Clip(006-015)-$Center$ MFI-$(4, 8)$
SV-CSTSR w/o FM	34.99/0.9292	35.03/0.9401
SV-CSTSR	**35.05/0.9295**	**35.12/0.9405**

TABLE VII

ABLATION STUDY OF THE MASK STRATEGY ON CLIP016-*Average*

Models	MFI-$(4, 3)$	MFI-$(4, 4)$	MFI-$(4, 5)$
SV-CSTSR w/o Mask	33.05/0.9380	32.90/0.9376	32.73/0.9370
SV-CSTSR	**33.21/0.9384**	**33.11/0.9381**	**33.01/0.9377**

TABLE VIII

ABLATION STUDY OF THE IDA AND TEBLOCK ON SFI-(4, 2) TASK

Models	IDA	TEBlock	Ave-Clip(001-005)
Model-I	×	✓	34.98/0.9291
Model-II	✓	×	35.01/0.9289
SV-CSTSR	✓	✓	**35.12/0.9405**

TABLE IX

ABLATION STUDY OF THE FS ON CLIP016-*Average*

Models	MFI-$(4, 4)$	MFI-$(6, 4)$	MFI-$(8, 4)$
SV-CSTSR w/o FS	33.00/0.9379	28.58/0.8512	26.60/0.7904
SV-CSTSR	**33.11/0.9381**	**28.71/0.8517**	**26.68/0.7928**

can be attributed to the denser coordinate grids and richer feature representations provided by larger N, which facilitate the reconstruction of finer details at larger scales. However, when N exceeds 3, the performance ceases to improve and experiences a slight decline. This degradation may stem from the limited parameter capacity of CFIM, making it difficult for the model to learn optimal parameters under the higher $N = 4$ setting. Considering the tradeoff between computational cost and reconstruction quality, we ultimately set $N = 3$ in our experiments.

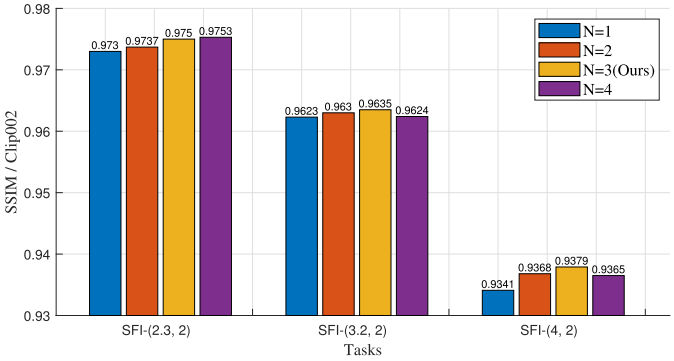


Fig. 13. Ablation experiments of the value N of CFIM on clip002 in terms of SSIM.

VI. CONCLUSION

In this article, we propose a novel satellite video CSTSR framework, termed SV-CSTSR. To effectively extract features of weak and complex textures in satellite videos, we develop an SFJMB, which can adaptively capture both spatial structural cues and frequency-domain texture patterns, thereby enhancing the feature representation of fine-grained details. To achieve continuous VFI, we propose an MTWM, which aims to use the time factor to dynamically adjust the deformable convolution offsets to achieve controllable feature warping. Furthermore, we design an IDAM to further refine the features of the interpolated video frame. Finally, we propose a CFIM to achieve continuous-scale upsampling. Specifically, our CFIM aims to learn an optimal FS strategy, which helps the network adaptively choose appropriate latent codes in the frequency domain to learn feature representations of satellite videos at arbitrary resolution. Benefiting from these innovations, SV-CSTSR achieves superior reconstruction quality compared with SOTA methods and significantly broadens the application scenarios of satellite videos.

In future work, on the one hand, we will explore more challenging scenarios, such as severe motion and heavy degradation, to further improve the robustness and generalization ability of the model. On the other hand, we aim to investigate more efficient network designs and lightweight formulations to reduce computational cost while maintaining high reconstruction quality, enabling practical deployment in real-time scenarios.

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